

Nash-Moser 迭代.

I. 导数损失 Loss of Derivatives

令 M^n 为一个 n 维流形, 且 $u: M^n \rightarrow \mathbb{R}^N$ 是一个到高维欧氏空间的嵌入.

M^n 上的坐标: $x = (x_1, \dots, x_n)$. $\partial_i u = \frac{\partial u}{\partial x_i} \in \mathbb{R}^N$

$$\Phi(u)_{ij} = \partial_i u \cdot \partial_j u$$

目标: 解

$$\Phi(u)_{ij} = \underbrace{f_{ij}}_{\text{对称}}, \quad 1 \leq i, j \leq n$$

$$\partial_i u \partial_j u =$$

· 线性化 Φ :

$$\begin{aligned} \Phi(u+v)_{ij} &= \partial_i(u+v) \partial_j(u+v) \\ &= \underbrace{\partial_i u \partial_j u}_{\text{已知}} + \underbrace{\partial_i u \partial_j v + \partial_i v \partial_j u}_{\text{线性}} + \cancel{\partial_i v \partial_j v}. \end{aligned}$$

$$(\Phi'(u)v)_{ij} = \partial_i u \partial_j v + \partial_i v \partial_j u.$$

要解

$$\Phi'(u)v = h$$

$$\partial_i u \partial_j v + \partial_j u \partial_i v = h_{ij}, \quad 1 \leq i, j \leq n$$

Nash: 假设正交条件:

$$\begin{cases} v \cdot \partial_k u = 0, & k=1, 2, \dots, n \\ \partial_i u \cdot \partial_j v + \partial_i v \cdot \partial_j u = h_{ij}, & 1 \leq i, j \leq n \end{cases} \quad \begin{matrix} (1) \\ (2) \end{matrix}$$

$$\partial_i(1) \Rightarrow$$

$$\partial_i(v \cdot \partial_k u) = 0$$

$$\Rightarrow \partial_i v \cdot \partial_k u + v \cdot \partial_i \partial_k u = 0$$

$$\Rightarrow \partial_i v \cdot \partial_k u = -v \cdot \partial_i^2 u$$

$$\text{代入 (2)} \Rightarrow -v \cdot \partial_i^2 j u - v \partial_j^2 i u = h_{ij}$$

$$\Rightarrow \begin{cases} -2v \cdot \partial_i^2 j u = h_{ij} \\ v \cdot \partial_i u = 0. \end{cases} \Rightarrow A(x) \cdot v(x) = b$$

$$A_{E \times N} \quad (E = ? = N)$$

$$\Rightarrow v(x) = A^{-1} b$$

$$\Rightarrow v(x) \propto \alpha \cdot \partial^2 u + \beta \cdot \partial u.$$

$$\text{有估计 } \|v\|_s \leq C(\|u\|_{s+2} \|h\|_0 + \|h\|_s),$$

I. Modifier

$$S_{\alpha n}(u) = \tilde{u}$$

Lemma 1. $\|(I - S_{\alpha n})e\|_S \leq C \cdot \alpha n^{-k}$, $k > 0$. k 依赖于初值

II. Nash-Moser Iteration

Def 1. (Tame Estimate, 柔性估计)

Let F, G be grade Fréchet space (Banach spaces) with norms $\|\cdot\|_s$. A non-linear mapping $\Phi: U \subseteq F \rightarrow G$ is tame, if $\forall s \geq b$, $\exists C_s$ s.t.

$$\|\Phi(u)\|_s \leq C_s (1 + \|u\|_{s+b})$$

解 $\Phi(u) = f$.

牛顿迭代: 在第 n 步, 对近似解 u_n , 残差是

$$e_n = f - \Phi(u_n)$$

目的: 找 h_n s.t. $u_{n+1} = u_n + h_n$.

由 Taylor 展开:

$$\Phi(u_n + h_n) = \Phi(u_n) + \Phi'(u_n)h_n + Q(u_n, h_n)$$

其中 $\Phi'(u_n)$ 是线性化算子,

$Q(u_n, h_n)$ 是余项 $O(\|h\|^2)$

标准牛顿迭代:

$$\Phi'(u_n)h_n = e_n \quad (\times)$$

Nash 迭代:

令 $v_n = S_{\alpha n}(u_n)$. 解

$$\Phi'(v_n)h_n = S_{\alpha n} e_n.$$

$$e_{n+1} = f - \Phi(u_{n+1}) \quad u_{n+1} = u_n + h_n$$

$$= f - \Phi(u_n + h_n)$$

$$= f - \Phi(u_n) - \Phi'(u_n)h_n - Q$$

$$= e_n - \Phi'(u_n)h_n - Q$$

$$= e_n - \Phi'(v_n)h_n + \Phi'(v_n)h_n - \Phi'(u_n)h_n - Q$$

$$= \underbrace{e_n - S_{\alpha n} e_n}_{\text{磨光误差}} + \underbrace{\Phi'(v_n)h_n - \Phi'(u_n)h_n}_{\text{替换误差}} - Q$$

磨光误差

替换误差.